

INVESTIGATION REPORT

Issue 1022: Arbitration between operation groups in the FPU

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1022
Disposition	Dead code, proven
Record date	2026-09-17
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1022
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL 6033d2b, COREV_PULP=1 FPU=1 (the upstream PULP_F0 configuration), top cv32e40p_formal_top, yosys-slang frontend
Disposition	Dead code, proven
Severity	Not assessed
Agreement	The human record stated no proof or test. This disposition agrees with everything the record did say and adds evidence.
Evidence	4 cited, 4 present

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The unreachable logic in the lzc instances follows from the arbiter above them never having more than one requester. The lzc pair the issue refers to lives in the gen_fair_arb block of rr_arb_tree and exists to choose between several simultaneous requesters. In this configuration the only rr_arb_tree instance with more than one input is the opgroup arbiter in fpnew_top, whose request vector is opgrp_out_valid, one bit per operation group. An unbounded proof (abc pdr, top cv32e40p_formal_top, COREV_PULP=1 FPU=1) shows \$countones(opgrp_out_valid) is never greater than 1: the core issues one APU operation at a time and waits for it, so two operation groups never present a result in the same cycle. The arbiter therefore never arbitrates, and the round-robin selection logic including the masked lzc is never exercised. A bounded cover run reaches each of the four operation groups individually, at steps 12, 14, 12 and 12, so all four requesters are live and the proof is not vacuous. This is the mechanism the issue describes as a one-core setup under-utilising the arbitration, stated as a property of the design rather than as an observation. The issue names no module instance and no line number, so this result identifies the cause; mapping it onto specific uncovered lines inside lzc.sv still needs the original coverage database.

3 Conditions

1. legal OBI handshakes on both the instruction and the data interface
2. scan_cg_en_i=0
3. no CSR write to the hardware-loop CSRs (0xCC0, 0xCC1, 0xCC2, 0xCC4, 0xCC5, 0xCC6)
4. clock gates modelled as transparent
5. every trace starts in reset
6. instruction memory contents otherwise unconstrained

Reproduced from the assumes that are enabled in the upstream formal environment at cores/cv32e40p/scripts/formal/src, plus the two modelling choices the open-source flow requires.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

CV32E40P FPU configuration, lzc instances inside rr_arb_tree gen_fair_arb: the round-robin selection path is unreachable because the arbiter never sees more than one requester. Formally proven (abc pdr, unbounded, top cv32e40p_formal_top, COREV_PULP=1 FPU=1, RTL 6033d2b) that \$countones(opgrp_out_valid) <= 1 at the fpnew_top opgroup arbiter, with all four operation groups shown individually reachable. Assumptions: legal OBI handshakes on both interfaces, scan_cg_en_i=0, no CSR write to the hardware-loop CSRs, transparent clock gates. Evidence: runs/v1-cv32e40p-1022/formal/p_assert/sby.log (PASS) and formal/c_antecedent/sby.log (4/4 requesters reachable). This establishes the cause. The specific uncovered lines inside lzc.sv are not enumerated here, because the issue does not name them and the coverage database was not available.

Checker used

checker_1022.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_assert/sby.log	present	abc pdr PASS (311.2 s, COREV_PULP=1 FPU=1): opgrp_out_valid never has more than one bit set, so the opgroup arbiter never has two requesters
E2	proof	formal/p_assert/result.json	present	machine-readable status record for the proof job
E3	proof	formal/c_antecedent/sby.log	present	smtbmc/yices cover PASS (103.8 s, depth 60): covers_reached=4 - each of the four operation groups is individually reachable, at steps 12, 14, 12 and 12, so every arbiter input is live
E4	proof	formal/c_antecedent/result.json	present	machine-readable status record for the cover job

Paths are relative to runs/v1-cv32e40p-1022. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.

7 Limitations and scope

- The proof establishes unreachability under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8

Provenance and measurement

Run directory	runs/v1-cv32e40p-1022
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024 (issue body): waived in v2; described as the LZC being under-utilised in a one-core setup. No assertion, module instance or line number given.
Assessed confidence	0.8. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	1

Measured tool and step time

Step	Measured
prove	5.2 min (311 s)
cover	1.7 min (104 s)
Total	6.9 min (415 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Generated by reporting/scripts/issue_document.py from runs/v1-cv32e40p-1022/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.